DESIGN COMPUTED · FREE GUIDE

Move the ground, not the tower: how Pisa was saved

The free guide to the reel on the Leaning Tower of Pisa: why it kept leaning, how engineers stopped it by removing soil, and how to see the same effect with a model.



Why the tower leans

The tower stands on soft, layered ground of clay and sand. The soil under its south side is a little more compressible than under the north side, so as the heavy stone tower was built, that side sank more and the tower began to tilt.

Once a tall, heavy object tilts, its weight moves towards the low side and pushes that side down even more. By the late twentieth century the lean was growing every year, and in 1990 the tower was closed to the public.

What engineers did

An international committee, chaired by Michele Jamiolkowski with John Burland among its members, took over. As a first, temporary step they placed lead counterweights at the foot of the north (high) side to slow the movement.

The lasting solution was soil extraction. From 1999 to 2001, drills slowly drew out small amounts of soil from under the north side, about 70 tonnes in all. The ground there settled slightly, and the tower rotated back by about 45 cm, to roughly where it stood in 1838. It reopened in 2001, still leaning, as everyone wanted.

Why it worked

Pushing or pulling the tower would have risked cracking its masonry, which was already highly stressed. Removing soil changes nothing in the stone: it simply lets the high side sink a few millimetres at a time, under the tower's own weight, while instruments watch every movement.

Built examples of acting on the ground

Leaning Tower of Pisa (1999-2001)	Controlled soil extraction under the high side; lean reduced by about 45 cm.
Metropolitan Cathedral, Mexico City (1990s)	Soil was also removed below parts of the cathedral to make uneven settlement more even (under-excavation).
Many sinking buildings worldwide	Foundations are underpinned or soil is grouted to stop movement, instead of rebuilding the structure above.

Try it yourself: a tower on sand

1. Fill a shallow tray with damp sand and level it. Push a heavy cylinder (a can full of coins) into it, slightly tilted, so it leans.
2. With a drinking straw or a thin spoon, carefully remove a tiny amount of sand from under the HIGH side, at an angle, without touching the can.
3. Wait a few seconds and tap the tray gently: the high side settles a little and the can straightens slightly.
4. Repeat in very small steps and measure the tilt each time with a phone level app.
5. Now try removing sand under the LOW side and watch the lean grow: where you act matters more than how much.

The rule to remember

When a structure is moving, the cheapest and safest fix is often below it: change what it stands on before you force the structure.

Default answer	Push, pull or brace the structure itself.
Form-finder answer	Let its own weight do the work by adjusting the ground beneath.
Check	Ask: is the problem in the building, or in what it stands on?

Go further

Leaning Tower of Pisa (Wikipedia)
https://en.wikipedia.org/wiki/Leaning_Tower_of_Pisa

Trevi: the restoration of the Leaning Tower of Pisa
https://trevispa.com/en/progetti/the-restoration-of-the-leaning-tower-of-pisa/

Construction Equipment Guide: Pisa leans on contractor to save its tower
https://www.constructionequipmentguide.com/pisa-leans-on-contractor-to-save-its-tower/9468

Credits
Cover photo: Saffron Blaze, CC BY-SA 3.0, Wikimedia Commons. Other photos: see the reel caption.
Facts: Wikipedia, Trevi S.p.A., Iowa State University (CE 203 report), Construction Equipment Guide.

Made by Design Computed · Follow for more · @designcomputed